

Section C – 1606001
Water (Overseas)

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Spec Item	Title	Description
1	General Information	
1.1	Outline of Services	The intent of 1606001 Water (Overseas) is to specify the requirements for the maintenance of raw water wells, water treatment plant instrumentation and control devices, inspection testing and certification of equipment, well redevelopment, and unilateral directional flushing. Service Order response for Water are included in 1601000 Utility Management.

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Spec Item	Title	Description
2	Management and Administration	
2.1	Definitions and Acronyms	Definitions and Acronyms provided in J-1606001-01.
2.2	Personnel	In addition to the Employee Requirements specified in Annex 2, the Contractor shall provide personnel with the qualifications, technical knowledge, experience and skills required to efficiently maintain the water treatment plant (WTP) instrumentation equipment and raw water wells.
2.2.1	Certification, Training and Licensing	The Contractor shall provide a subject matter expert (SME) available or on call qualified to troubleshoot any system programmable logic controls (PLCs), controls, and instrumentation failures. The Contractor shall submit proof of all certification, training, and licensing requirements per Section F.
2.3	Special Requirements	
2.3.1	System Certifications	The Contractor shall support all inspection, survey, data review, and corrective action update status efforts, associated with the water systems' Certificate to Operate (CTO) review process specified in CNIC M-5090.1. These efforts include, but are not limited to, the Sanitary Survey, Request for Information (RFI), and updates to the Requirements POAM.
2.3.2	Water Systems Equipment	Water instrumentation equipment shall be certified and calibrated annually by an approved certifying office. Waste equipment shall be calibrated and certified in accordance with UNI EN ISO / IEC 17025. Calibration certificates shall be accredited by ACCREDIA. Calibration reports shall be developed and provided in accordance with ISO 9001, per Section F.
2.4	References and Technical Documents	References and Technical Documents provided in J-1606001-02.

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3	Recurring Work	The Contractor shall maintain the raw water supply wells, water treatment plant instrumentation equipment and systems, and unilateral flushing of the distribution systems to ensure the safe production, treatment and distribution of quality, reliable drinking water such that all drinking water demand requirements are continually met.	The Contractor shall maintain all water systems in accordance with the most stringent standards specified by Italian and U.S. laws and regulations, specifically FGS Italy, CNIC M-5090.1, ASME B40.100-2005, FGS Italy C3.3.2.7, and UFC 3-230-02, Ch. 8 (8-3), equipment manufacturer's instructions or recommendations, and industry standards, and references and technical documents. Conflicting issues shall be brought to the immediate attention of the KO. The Contractor shall shut-down, restart, and perform operational checks on all equipment affected by both scheduled and unscheduled utility outages as requested and/or directed by PWD UEM/EV personnel at no additional cost to the Government. The Contractor shall coordinate all maintenance and repair scheduled outages in a manner that minimizes the disruption of water services. Outage requests for maintenance and repairs that will cause disruption of water services shall be submitted per Section F and must be approved by KO five (5) workdays prior to commencement of work. The Contractor shall provide verbal notification to PWD UEM representative within 10 minutes of identification of any unscheduled disruption to the water systems, and also provide an Unscheduled Water Disruption Situation Report per Section F to document the cause and follow-up actions to unscheduled disruptions to the water system's capability to continually meet FFHC and non-FFHC water demands and	Fit for Human Consumption (FFHC) and non-FFHC water is continuously and safely produced and distributed and meets quality standards and demand requirements. Critical spare parts inventory is current and available 100% of the time.

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			continually maintain water quality per specified criteria. The Contractor shall accommodate and support the Government's periodic inspection, survey, and data review efforts associated with the water systems' Certificate to Operate (CTO) and sanitary survey review process specified in CNIC M-5090.1. The Water Systems Descriptions and Inventories are provided in J-1606001-03. The Contractor shall develop and submit an inventory of Critical Spare Parts per Section F. The Contractor is fully responsible for maintaining on hand the critical spare parts required to keep the system in continuous service to meet water distribution requirements. The Historical Critical Spare Parts Inventory can be found in J-1606001-04. The historical list is provided as information only and is not to be construed as complete and adequate. The Contractor is fully responsible for ensuring the critical spare parts list is updated and expanded as required and parts are available at all times. The Government will inspect maintained spare parts quarterly to ensure adequate storing and ordering levels.	
3.1	Preventive Maintenance (PM) Program	The Contractor shall develop and implement a Preventive Maintenance (PM) program for the raw water supply wells and water treatment plant	The Contractor is fully responsible for and shall perform any repairs, including replacement, discovered during scheduled maintenance work up to a total of \$3,000 per occurrence in direct material and labor cost under the recurring work portion of the contract.	Maintenance accomplished in accordance with the Contractor's PM program and work schedule. PM performed in accordance with

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		instrumentation equipment and systems to ensure proper operation, to minimize breakdowns, and to maximize useful life.	Incidental repairs work performed under maintenance are not considered a service order. Notification of repair work exceeding the incidental repairs limit shall be submitted to the KO within two hours of identification per Section F. Service orders or non-recurring work task orders may be issued for repairs exceeding the incidental repairs limit. The Contractor shall not use breakdown maintenance as part of the PM program. The PM program shall provide an economical approach manufacturers' recommended procedures, OEM standards, and maintenance required to satisfy equipment warranties and keep the water plant and distribution system equipment in normal working condition. Excessive or repeated system or equipment breakdowns or deficiencies may indicate the need to adjust or modify the Contractor's PM program. These changes shall be made at no additional cost to the Government. If the KO determines that the objective and standards are not met, the Contractor shall adjust or modify the Contractor's PM program at no additional cost to the Government. The Raw Water Supply, Water Treatment Plant and Distribution Systems Descriptions and Inventories are provided in Section J. The Contractor shall submit a monthly maintenance schedule,	manufacturers' recommended procedures, UICAP requirements, OEM standards, latest editions of the AWWA Manual of Water Supply Practices and UFC 3-230-02. Annual raw water collection system cleaning and disinfection performed per specified SOP and requirements.

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			unaccomplished maintenance report, repair status report, and summary of maintenance and repair accomplished under the various maintenance programs report per Section F. The Contractor shall submit separate monthly reports for: • Annual maintenance schedule • Monthly maintenance schedule • Unaccomplished maintenance report • Repair status report per Section F.	
3.1.1	Calibration and Maintenance of Water System Instrumentation and Equipment	The Contractor shall perform calibration and maintenance on water systems instrumentation equipment to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall calibrate water instrumentation and equipment located at all water system sites. Calibration shall be performed annually by certified operators. Any instrumentation equipment that is beyond repair shall be replaced with new equipment matching the features and capabilities of the replaced equipment. The Contractor shall test replacement equipment for compliance with the installed PLCs and SCADA. The contractor shall submit reports detailing results of calibrations per Section F. The Contractor shall submit as well as technical data sheets of new items installed and a year warranty for any manufacturing defect, improper installation and/or malfunctioning. Calibration and maintenance shall be performed in accordance manufacturers' recommended procedures and OEM standards. The Contractor shall develop and submit, per Section F, a Water System Instrumentation	Calibration and maintenance are performed in accordance with Contractor's PM program and work schedule. Testing, calibration and maintenance are performed in accordance with manufacturers' recommended procedures and OEM standards. Instrumentation and control devices are calibrated, accurate, and operable.

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			and Control Devices' Testing and Calibration Schedule per manufacturers' requirements, recommendations and frequencies. The Contractor shall repair or replace instrumentation and control devices that cannot be properly calibrated or function as intended. The Contractor shall notify KO immediately with instrumentation and control issues that can potentially affect safe, effective, and accurate water operations. All instrumentation and control devices, including replacements, shall be capable of being remote read per the Navy Utility Metering Policy.	
3.1.1.1	Probes and Sensors	The Contractor shall perform calibration and maintenance on water systems probes and sensors to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall calibrate all probes and sensors at the NAS 1 and NAS 2 WTPs. Probes and sensors include, but are not limited to, flow rate sensors, on line chemical analyzer probes, RO conductivity, pH, ORP, Chlorine, and turbidity probe. The Contractor shall calibrate equipment annually. The Contractor shall replace any probe or sensor beyond repair with new equipment with the same features and capabilities as the replaced equipment. The Contractor shall test all replacements for compliance with installed PLCs and SCADA. The Contractor shall submit reports detailing results of calibrations per Section F. The Contractor shall submit as well as technical data sheets of new items installed and a year warranty for any manufacturing defect, improper installation and/or malfunctioning.	Work is performed in accordance with Contractor's PM and work schedule.

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3.1.1.2	Pressure Gauges	The Contractor shall perform calibration and maintenance on water systems pressure gauges to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall calibrate all pressure gauges at the NAS 1, NAS 2 and Niscemi WTPs and water system locations. Pressure gauges include, but are not limited to, the pressure gauges installed in the RO rooms, wells, raw water tanks, TTW tanks, multimedia filters, calcite contactors and buffer tanks. Calibration shall be performed annually by certified operators. Any pressure gauge beyond repair shall be replaced with a new one matching the same features and capabilities of the one being replaced. The Contractor shall submit reports detailing results of calibrations per Section F. The Contractor shall submit as well as technical data sheets of new items installed and a year warranty for any manufacturing defect, improper installation and/or malfunctioning.	Work is performed in accordance with Contractor's PM and work schedule.
3.1.2	Raw Water Wells	The Contractor shall perform PM for raw water wells to ensure safe, reliable, uninterrupted fire protection service.	The Contractor shall perform annual preventative maintenance on wells, well pumps and equipment, and well backwash in accordance with standards and regulations set by the American Water Works Association (AWWA), NAVFAC UEM SOPs, and EPA recommendations and requirements. As part of the PM, the Contractor shall perform complete well cleaning for each well. The Contractor shall conduct video inspections of each well every six months. The Contractor shall provide video inspection results and findings per Section F.	Work is performed in accordance with Contractor's PM and work schedule. Video inspection results and findings submitted as required.
3.1.3	Fire Water Pump Stations	The Contractor shall perform a PM	The Contractor shall perform maintenance to fire water pump	Work is performed in accordance with

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		for fire water pump stations to ensure safe, reliable, uninterrupted fire protection service.	station equipment. Work includes equipment located at the flight line fire main, the fuel compound, the PWD compound, and the NAVSUP/DLA warehouse compound. The PM plan shall comply with established guidelines in UFC 3-601-02, Operations and Maintenance: Inspection, Testing, and Maintenance of Fire Protection Systems. Repairs shall meet all OEM requirements and NFPA codes and standards. The Contractor shall notify facility occupants, Fire Department, and Security Department 24 hours and immediately prior to performing maintenance, inspection, or testing of fire suppression systems. The Contractor shall notify the Fire Department within one hour upon discovery of any malfunctioning system or equipment. Fire Main Systems Map is provided in J-1606001-05.	Contractor's PM and work schedule.
3.2	Unidirectional Flushing (UDF) Plan	The Contractor shall provide unidirectional flushing of water pipelines to ensure water distribution pipelines are cleaned.	The Contractor shall perform annual drinking water unidirectional flushing in compliance with standards and regulations set by American Water Works Association (AWWA) and National Fire Protection Association (NFPA) for Installation, Field Flow Testing, and EPA recommendations and requirements. The Contractor shall perform unidirectional flushing annually of the entire inventory of the pipelines at each location, NAS I, NAS II, and Niscemi. The	Work is performed in accordance with Contractor's PM and work schedule.

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			Contractor shall comply with the Unidirectional Flushing Plan procedures are provided in J-1606001-06 for NAS I and J-1606001-07 for NAS II. The Contractor shall submit a UDF plan detailing the dates and time flushing will take place by location across the entire water pipeline system per Section F.	
3.3	Well Development and Water Treatment and Distribution System Support	The Contractor shall provide labor and equipment support for well development and water distribution systems to ensure safe, reliable, uninterrupted water supply.	The Contractor shall perform labor and equipment services in support of in-house maintenance and repair work across the water system. Work may require the use of cranes/forklifts to lift and reinstall submersible pumps and/or and motors installed below ground. The Contractor shall be fully responsible for providing cranes and forklift equipment and operators necessary to support the work. The Contractor shall provide support for at least one well development effort annually. Each well development effort will require up to one week of support services. Informational Note: In addition to the once per year well development support, the Contractor can expect to support up to X projects across the water system annually.	Work is performed in accordance with Contractor's work schedule.

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4	Non-Recurring Work	Non-recurring work may be ordered utilizing DoD FedMall in accordance with Section H or on a task order in accordance with the PROCEDURES FOR ISSUING ORDERS clause in Section G. The order will specify the exact locations and types of work to be accomplished. The period of performance will be specified in each order.	Refer to non-recurring work ELINs for task listings, descriptions and related requirements. All periods of performance are measured from issue date of order to acceptance of the work.	Performance Standards for non-recurring work will be the same as those in Spec Item 3 where applicable.